

ZIDANE VIRANI

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Education

Western University

Apr 2028

Bachelor of Engineering in Software Engineering

- *Courses: Data Structures & Algorithms, Operating Systems, Parallel Computing, Probability & Statistics, Statistical Learning, Embedded Systems, Theory of Computation, Deep Learning, Computer Architecture, Neural Networks, Reinforcement Learning*
- *Involvements: Director at Western AI, Academic Mentor for Freshmen, Outreach chair of Western Dental Association*

Experience

Bloomberg LP: Software Engineer Intern

May 2026 – Aug 2026

- Parallelized a single-threaded C++17 job across a 300K+ bond universe through multithreading, rebuilding batch execution as a std::thread worker pool with std::mutex-guarded work queues, cutting runtime from 500+ hours to 30 minutes (~1000×)
- Redesigned performance-critical stages of a C++17 analytics pipeline after profiling with Linux perf and Google Benchmark, improving end-to-end throughput by 140% by eliminating redundant memory allocations, and repeated lookups
- Built C++ validation and test coverage for pricing and screening logic with GoogleTest and CMake, increasing market-data edge-case coverage by 35% and making future code changes easier to verify

Shopify: Applied Machine Learning Engineer Intern

Jan 2026 - Apr 2026

- Built a Python and SQL training-data pipeline for Shopify Sidekick, using Polars and NumPy to clean 2M+ merchant-query logs and generate features, labels, and evaluation datasets for retraining workflows
- Reworked Sidekick's feature generation pipeline with Python, Polars and NumPy, replacing slower Pandas workflows and reducing preprocessing runtime by 60% for ML experiments
- Optimized Sidekick's LLM inference by exporting PyTorch models to ONNX, improving serving speed and GPU efficiency

Western University: Machine Learning Researcher Intern

Jun 2025 – Dec 2025

- Accelerated PyTorch video-sequence model training on 120+ hours of debate footage, reducing per-epoch runtime by 45% through GPU training optimizations done with CUDA in C++ and parallel data loading
- Built a Python pipeline using OpenCV and NumPy to decode 10M+ video frames, sync them with transcripts, and label which speaker was talking in each segment for model training
- Trained a ResNet-50 + temporal Transformer end to end in PyTorch, hitting 98.86% validation accuracy identifying active speakers across 200K+ clips of unscripted debate footage

Vector Institute: Software Engineer Intern

Apr 2025 - Jun 2025

- Built a low-latency Python-based FastAPI inference service for real-time speech recognition, using async request handling and TensorFlow workers to handle 50 concurrent audio requests at 850ms p95 latency
- Developed Python evaluation tools with NumPy, Polars, and scikit-learn to flag mislabeled voice commands and low-confidence predictions across 20K+ audio examples, improving command classification accuracy by 38% after retraining
- Trained TensorFlow voice-command classification models to reach 99.96% accuracy on test dataset with 200k+ audio samples

Projects & Awards

Founder & Sole Engineer, AI Content Engine (Acquired for \$50K) ([Blog Link](#))

Jan 2023 – Dec 2024

- Built and sold a Python-based ranking and retrieval platform, scaling a profitable software product to 120K+ monthly readers and \$2,800/month in net profit through search, recommendation, and user-feedback systems
- Engineered Python and SQL data pipelines to process reader behavior, article metadata, and search activity, training a PyTorch ranking model on 10,000+ labeled examples for article recommendation and ranking
- Optimized retrieval, scoring, and evaluation pipelines through SQL query tuning, caching, and batching, reducing compute costs by 17% while improving recommendation iteration speed under production traffic

IBM Research ([Award Link](#)) | C++, STL, Algorithms, Optimization

Jun 2026

- Developed a high performance C++17 solver for an IBM Research public algorithms challenge, using STL containers, custom state encoding, and pruning to efficiently search an exponential solution space and earn recognition on IBM Research's website

Leadership

Aga Khan Development Network: Chair & Lead Engineer

Feb 2026 – Present

- Led a 6-person team to build and deploy a fair-wage matching platform in India, designing FastAPI backend services, AWS deployment infrastructure, and recommendation workflows that secured \$100K in funding and supported 2,500+ users

Technical Skills

- Languages: C++17/20, C, Python, SQL, Bash
- Systems & Tools: STL, Boost, CMake, Linux, Unix, Git, GDB, Linux perf, GoogleTest, Google Benchmark, PyTest
- Data & ML: PyTorch, CUDA, ONNX, NumPy, Polars, Pandas, Docker, AWS, scikit-learn, OpenCV, Redis